Chapter 1. INTEGRATION

(Page 5-106, Calculus Volume 1,2)



MAE101

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INTEGRATION

1.1

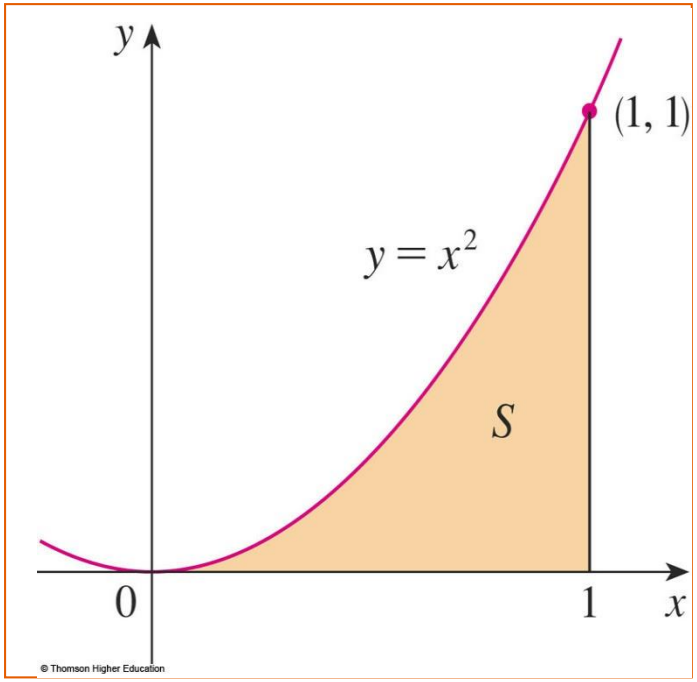
Approximating Areas

In this section, we will learn that:

We get the same special type of limit in trying to find the area under a curve.

AREA PROBLEM

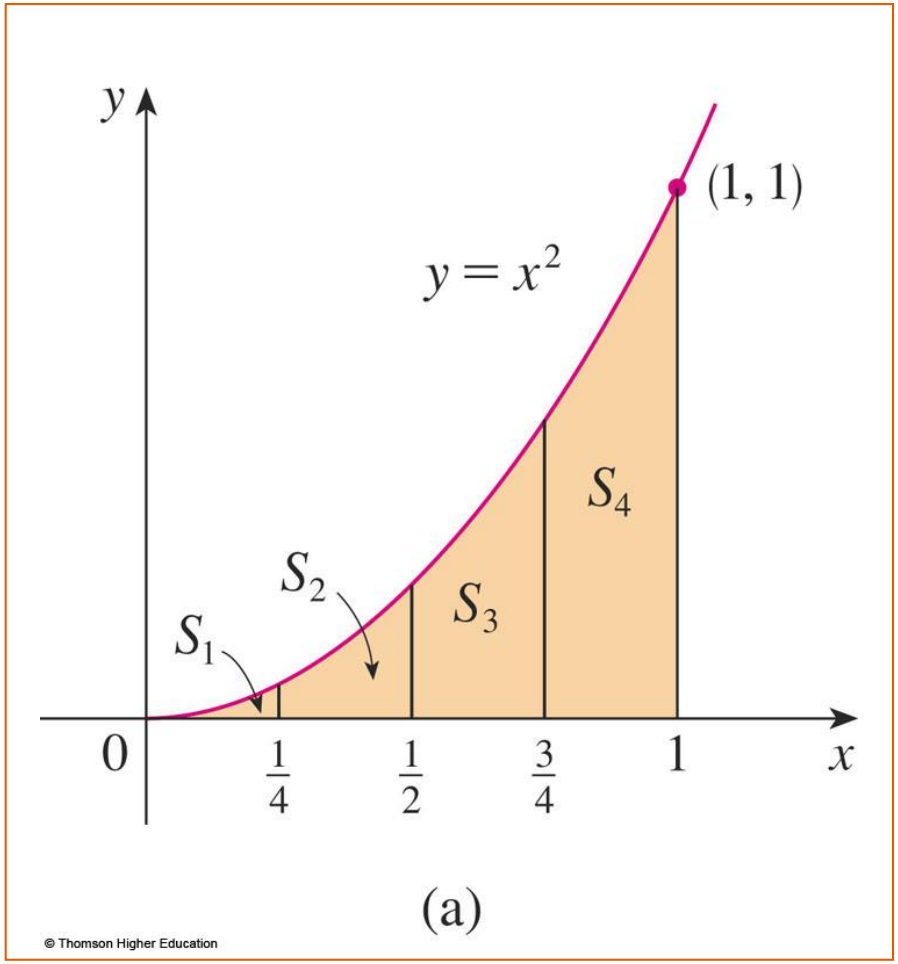
We begin by attempting to solve the area problem:
Find the area of the region S that lies under the curve $y = f(x)$ from a to b.



AREA PROBLEM

Example 1

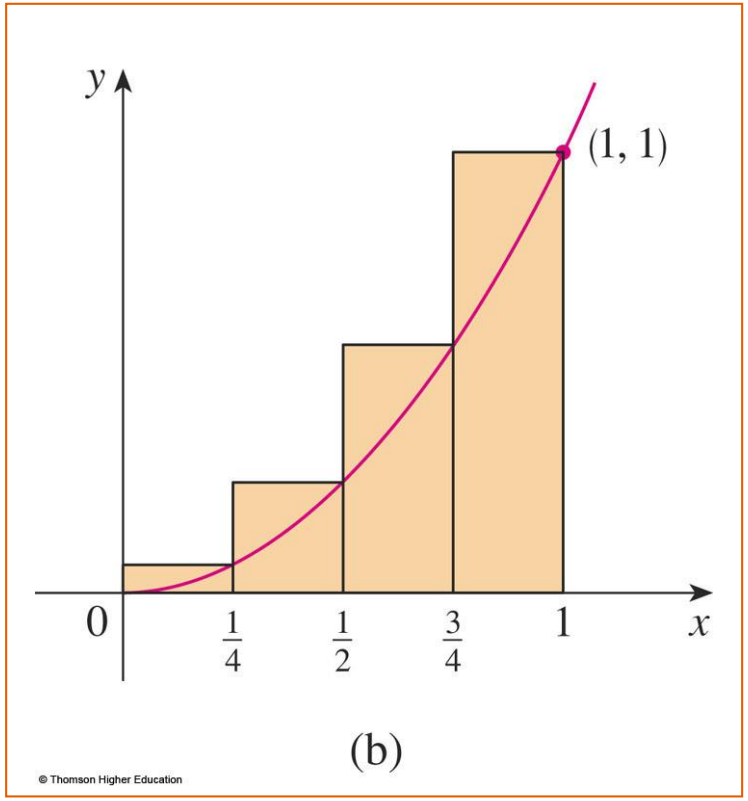
Suppose we divide S into four strips S_1, S_2, S_3, and S_4 by drawing the vertical lines $x = ¼$, $x = ½$, and $x = ¾$.



AREA PROBLEM

Example 1

The heights of these rectangles are the values of the function $f(x) = x^2$ at the **right endpoints** of the subintervals $[0, ¼], [¼, ½], [½, ¾]$, and $[¾, 1]$.



AREA PROBLEM

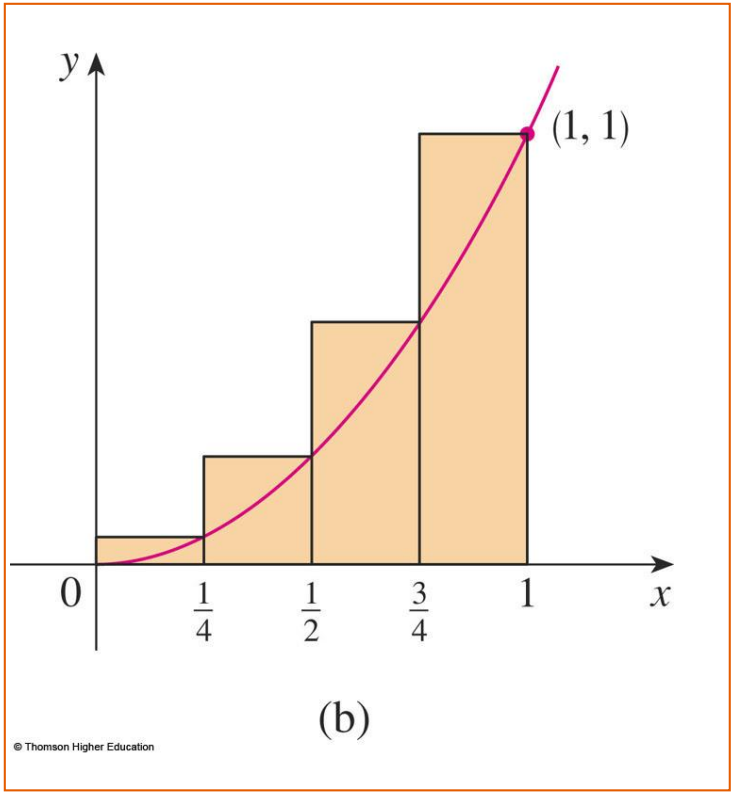
Example 1

Each rectangle has width ¼ and the heights are $(¼)^2$, $(½)^2$, $(¾)^2$, and 1^2.

$$R_4 = \frac{1}{4}\cdot\left(\frac{1}{4}\right)^2 + \frac{1}{4}\cdot\left(\frac{1}{2}\right)^2 + \frac{1}{4}\cdot\left(\frac{3}{4}\right)^2 + \frac{1}{4}\cdot 1^2$$

$$= \frac{15}{32}$$

$$= 0.46875$$



AREA PROBLEM

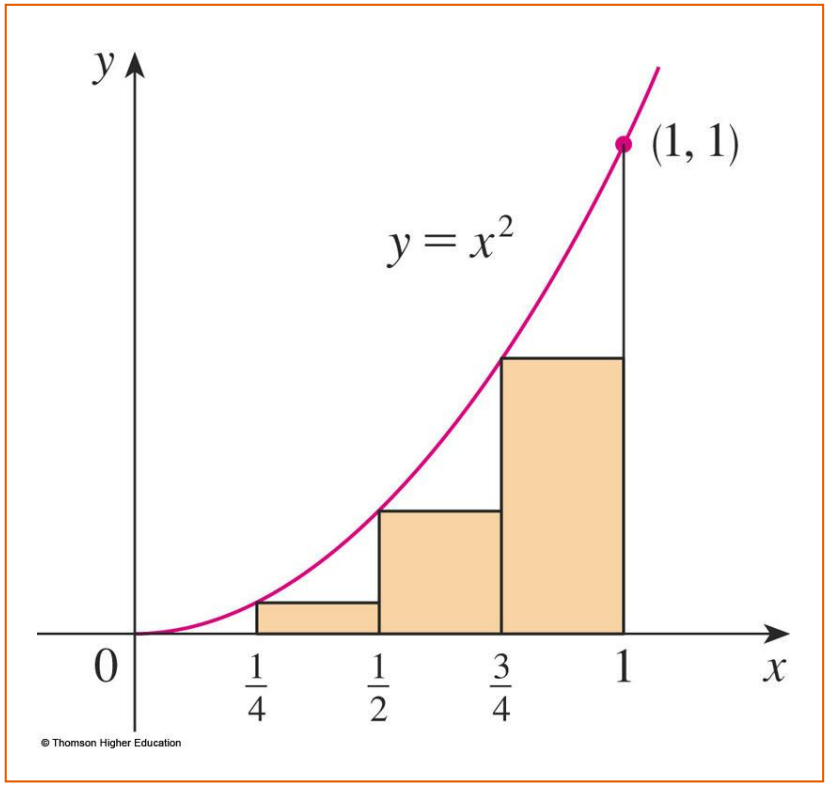
Example 1

Here, the heights are the values of f at the **left endpoints** of the subintervals.

$$L_4 = \frac{1}{4}\cdot 0^2 + \frac{1}{4}\cdot\left(\frac{1}{4}\right)^2 + \frac{1}{4}\cdot\left(\frac{1}{2}\right)^2 + \frac{1}{4}\cdot\left(\frac{3}{4}\right)^2$$

$$= \frac{7}{32}$$

$$= 0.21875$$

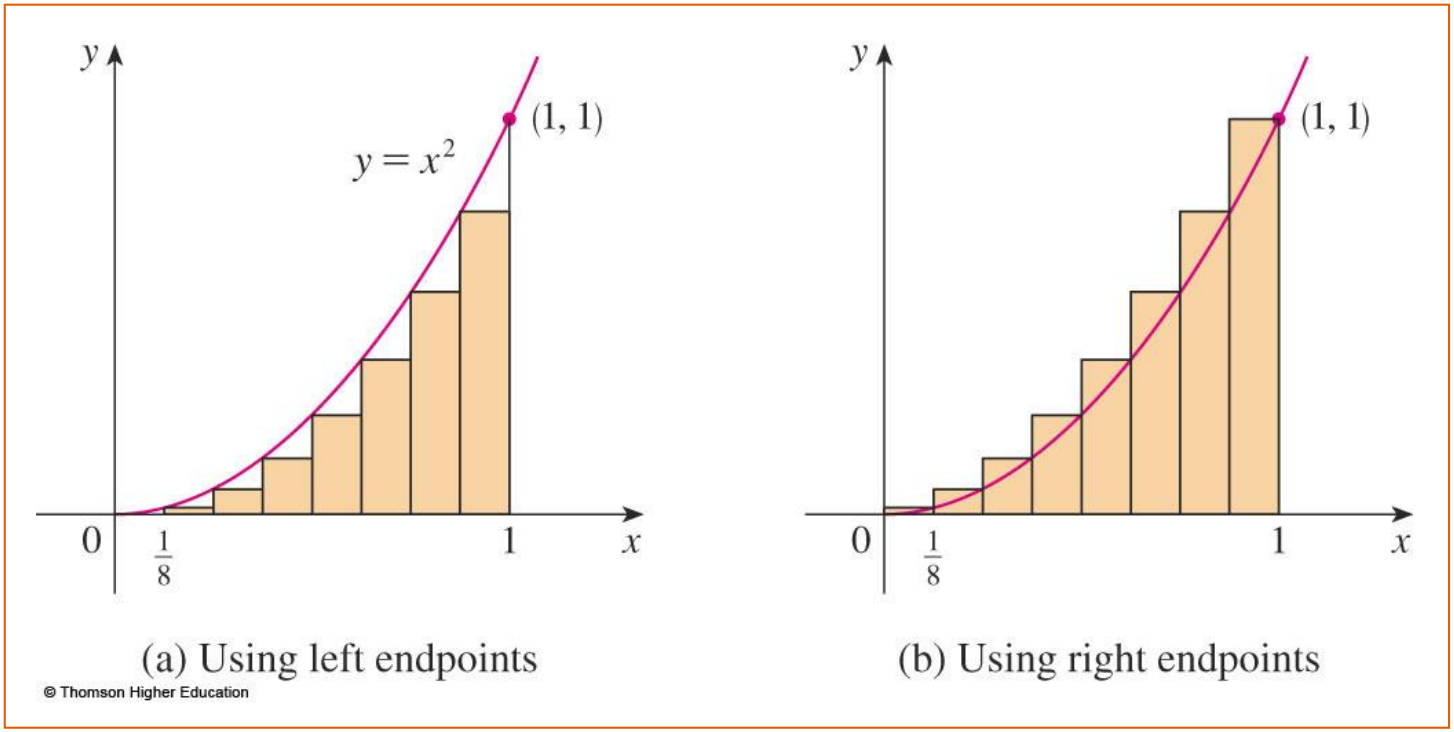


AREA PROBLEM

Example 1

The figure shows what happens when we divide the region S into eight strips of equal width.

$$0.2734375 < A < 0.3984375$$



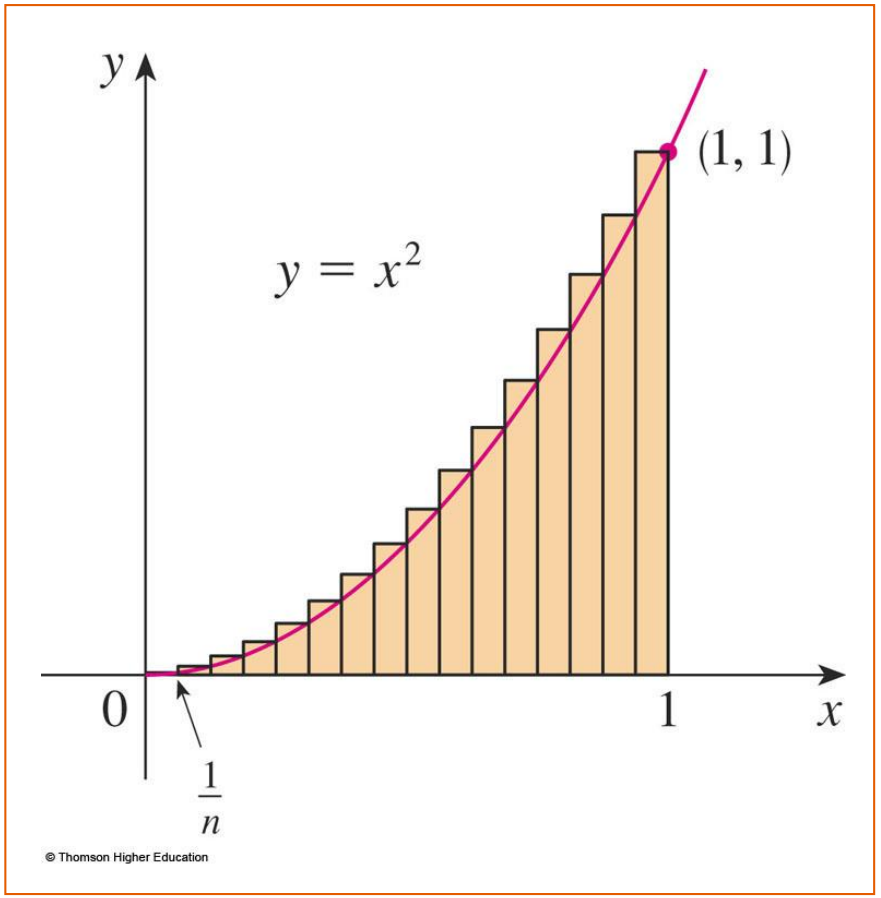
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AREA PROBLEM

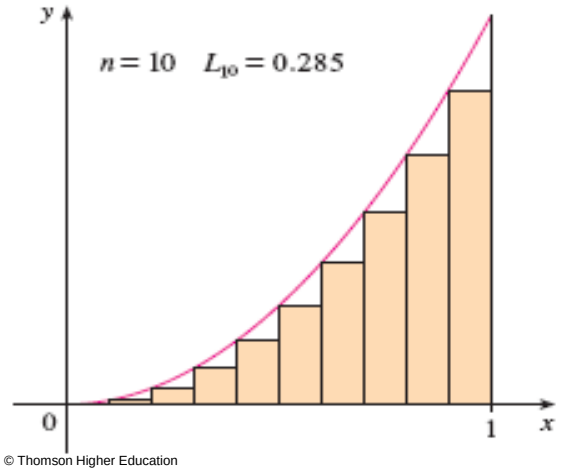
Example 2

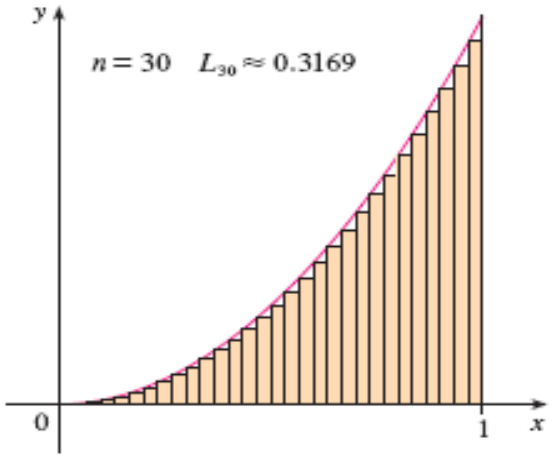
R_n is the sum of the areas of the n rectangles.

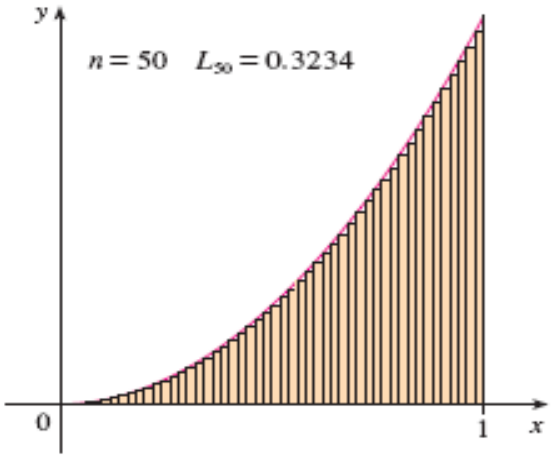
- Each rectangle has width $1/n$ and the heights are the values of the function $f(x) = x^2$ at the points $1/n, 2/n, 3/n, \ldots, n/n$.
- That is, the heights are $(1/n)^2, (2/n)^2, (3/n)^2, \ldots, (n/n)^2$.



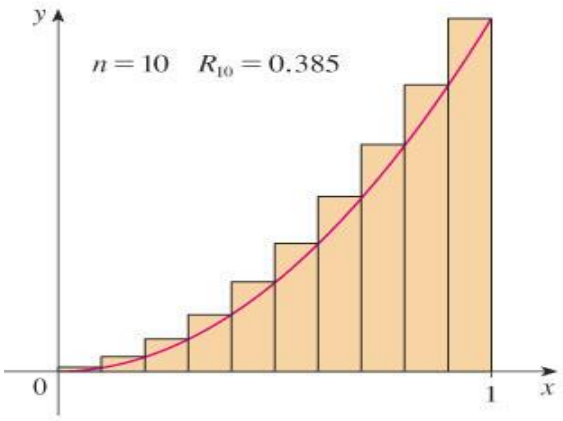
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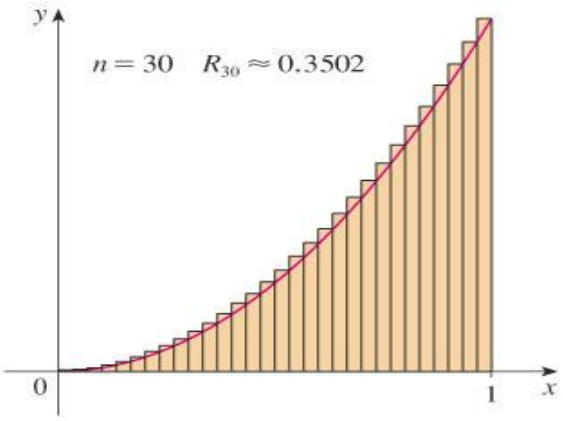


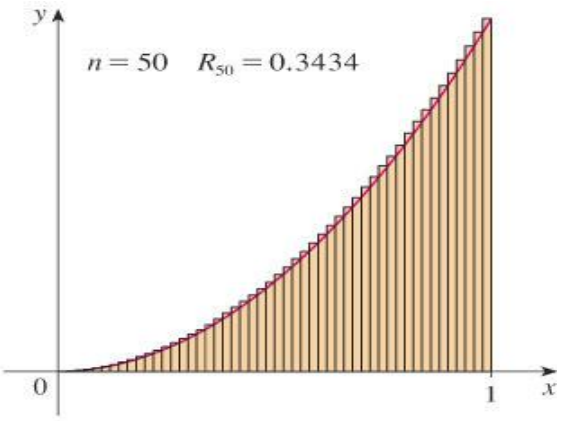




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AREA PROBLEM

Example 2

Thus,

$$R_n = \frac{1}{n}\left(\frac{1}{n}\right)^2 + \frac{1}{n}\left(\frac{2}{n}\right)^2 + \frac{1}{n}\left(\frac{3}{n}\right)^2 + \ldots + \frac{1}{n}\left(\frac{n}{n}\right)^2$$

$$= \frac{1}{n^3}(1^2 + 2^2 + 3^2 + \ldots + n^2)$$

$$R_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}$$

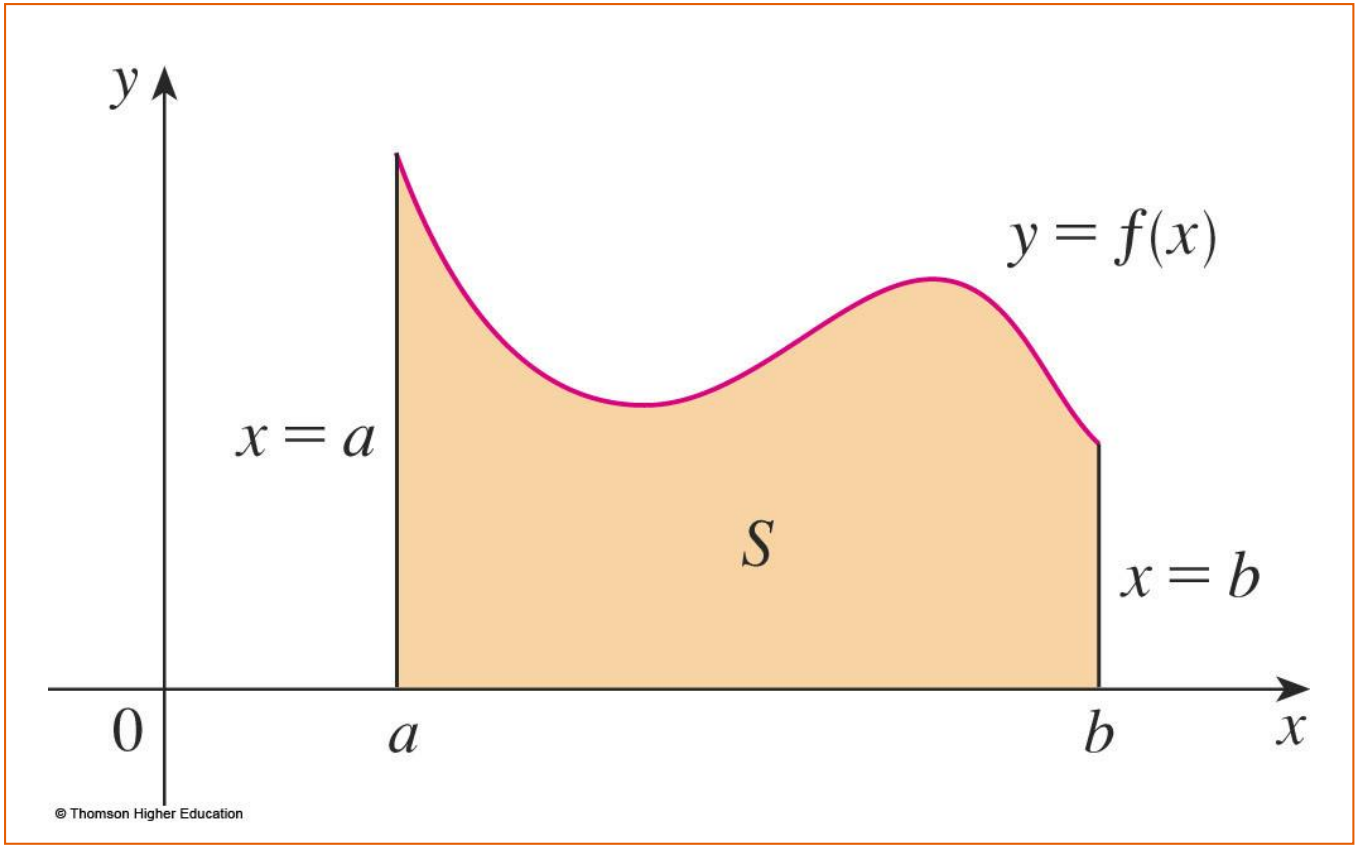
AREA PROBLEM

Thus, we define the area A to be the limit of the sums of the areas of the approximating rectangles, that is,

$$A = \lim_{n\to\infty} R_n = \lim_{n\to\infty} L_n = \frac{1}{3}$$

AREA PROBLEM

Let's apply the idea of Examples 1 and 2
to the more general region S of the earlier figure.

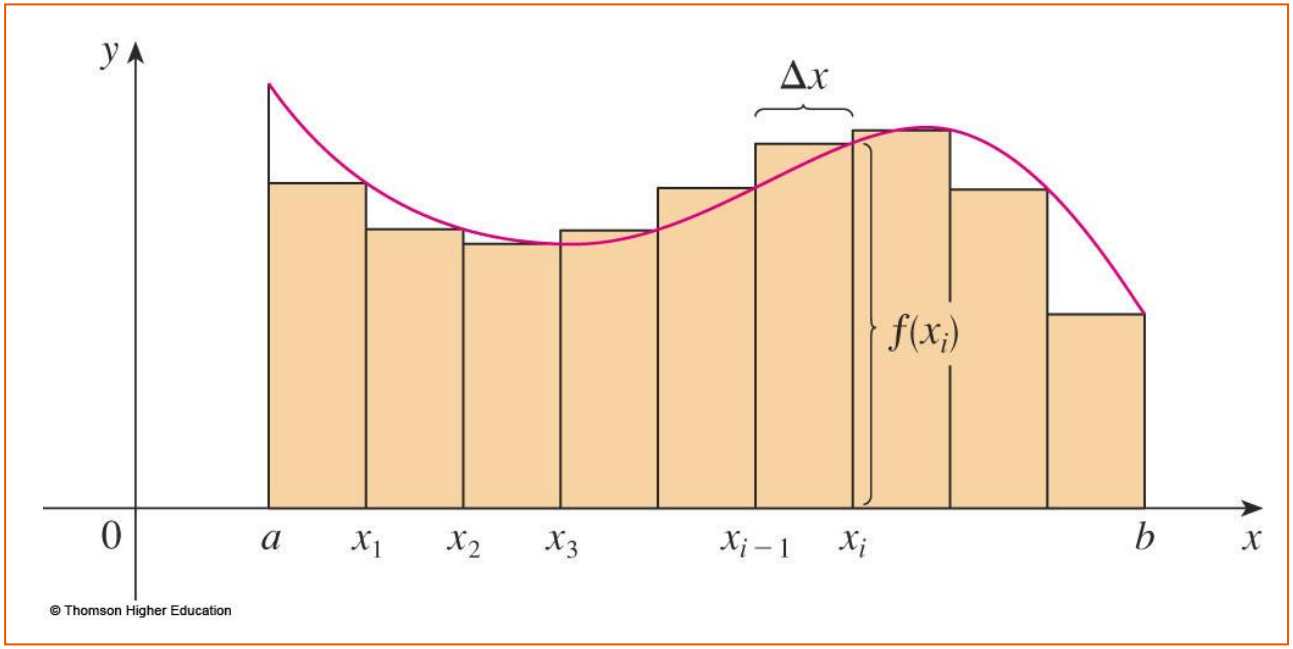


AREA PROBLEM

What we think of intuitively as the area of S is approximated by the sum of the areas of these rectangles:

$$R_n = f(x_1)\,\Delta x + f(x_2)\,\Delta x + \ldots + f(x_n)\,\Delta x$$

$$\Delta x = \frac{b-a}{n}$$



AREA PROBLEM

The area A of the region S that lies under the graph of the continuous function f is the limit of the sum of the areas of approximating rectangles:

$$A = \lim_{n\to\infty} R_n$$
$$= \lim_{n\to\infty}[\, f(x_1)\Delta x + f(x_2)\Delta x + \ldots + f(x_n)\Delta x]$$

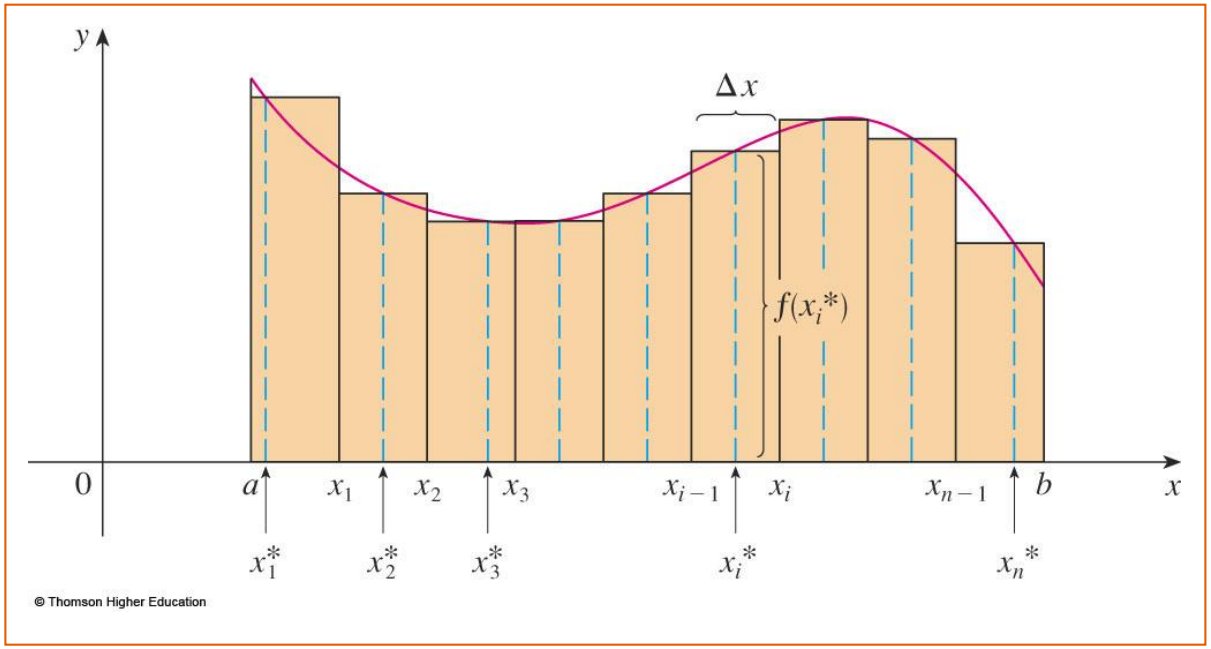
$$A = \lim_{n\to\infty} L_n$$
$$= \lim_{n\to\infty}[\, f(x_0)\Delta x + f(x_1)\Delta x + \ldots + f(x_{n-1})\Delta x]$$

AREA PROBLEM

The figure shows approximating rectangles when the sample points are not chosen to be endpoints.

x_i^* in the ith subinterval $[x_{i-1}, x_i]$

(the sample points)



$$A = \lim_{n\to\infty}[\, f(x_1^*)\Delta x + f(x_2^*)\Delta x + \ldots + f(x_n^*)\Delta x]$$

Hence,

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$$

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_{i-1})\Delta x$$

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

The sum $\sum_{i=1}^{n} f(x_i^*)\Delta x$ is called a Riemann sum.

Example

1. Evaluating the Riemann sum of the following function with four subintervals. The sample points are the **right endpoints.**

   f(x)= x-1/x (1<=x<=2)

2. **Estimate the area of the region under the graph of y = sin x on [0,5] with 3 subintervals [0,1]; [1,2.5] and [2.5,5] by using left-endpoints**

3. **Find the lower sum for $f(x) = 10 - x^2$ on [1,2] with 4 subintervals**

INTEGRATION

1.2

The Definite Integral

In this section, we will learn about:

Integrals with limits that represent

a definite quantity.

DEFINITE INTEGRAL

If f is a function defined for $a \le x \le b$,
we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$.

- We let $x_0 (= a), x_1, x_2, \ldots, x_n (= b)$ be the endpoints of these subintervals.

- We let $x_1^*, x_2^*, \ldots, x_n^*$ be any sample points in these subintervals, so x_i^* lies in the ith subinterval.

DEFINITE INTEGRAL

Then, the definite integral of *f* from *a* to *b* is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

provided that this limit exists.

☞ If it does exist, we say *f* is integrable on [*a*, *b*].

☞ The symbol $\int$ was introduced by Leibniz and is called an integral sign.

The definite integral $\int_a^b f(x)dx$ is a number.

It does not depend on x.

In fact, we could use any letter in place of x without changing the value of the integral:

$$\int_a^b f(x)dx = \int_a^b f(t)dt = \int_a^b f(r)dr$$

DEFINITE INTEGRAL

Definition

Let $f(x)$ be an integrable function defined on an interval $[a, b]$. Let A_1 represent the area between $f(x)$ and the x-axis that lies *above* the axis and let A_2 represent the area between $f(x)$ and the x-axis that lies *below* the axis. Then, the **net signed area** between $f(x)$ and the x-axis is given by

$$\int_a^b f(x)dx = A_1 - A_2.$$

The **total area** between $f(x)$ and the x-axis is given by

$$\int_a^b |f(x)|dx = A_1 + A_2.$$

INTEGRABLE FUNCTIONS

If *f* is continuous on [*a*, *b*], or if *f* has only a finite number of jump discontinuities, then *f* is integrable on [*a*, *b*].

That is, the definite integral $\int_a^b f(x)\,dx$ exists.

INTEGRABLE FUNCTIONS

If f is integrable on $[a, b]$, then

$$\int_a^b f(x)\,dx = \lim_{n_i \to \infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

where

$$\Delta x = \frac{b-a}{n} \quad \text{and} \quad x_i = a + i\,\Delta x$$

EVALUATING INTEGRALS

The following three equations give formulas for sums of powers of positive integers.

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$

PROPERTIES OF THE INTEGRAL

We assume f and g are continuous functions.

1. $\int_a^b c\,dx = c(b-a)$, where c is any constant
2. $\int_a^b [f(x)+g(x)]dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$
3. $\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$, where c is any constant
4. $\int_a^b [f(x)-g(x)]dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$

COMPARISON PROPERTIES OF THE INTEGRAL

These properties, in which we compare sizes of functions and sizes of integrals, are true only if $a \le b$.

6. If $f(x) \ge 0$ for $a \le x \le b$, then $\int_a^b f(x)\,dx \ge 0$

7. If $f(x) \ge g(x)$ for $a \le x \le b$, then $\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$

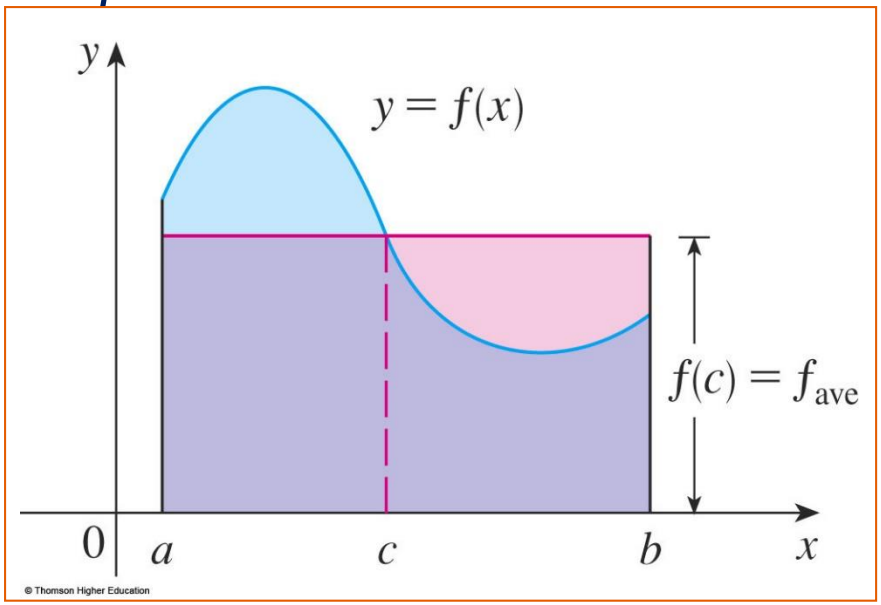
8. If $m \le f(x) \le M$ for $a \le x \le b$, then

$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a)$$

AVERAGE VALUE OF A FUNCTION

The geometric interpretation of the Mean Value Theorem for Integrals is as follows.

– For 'positive' functions f, there is a number c such that the rectangle with base $[a, b]$ and height $f(c)$ has the same area as the region under the graph of f from a to b.



AVERAGE VALUE OF A FUNCTION

If *f* is continuous on [*a*, *b*], then there exists a number *c* in [*a*, *b*] such that

that is, $$f(c) = f_{ave} = \frac{1}{b - a}\int_a^b f(x)\,dx$$

$$\int_a^b f(x)\,dx = f(c)(b - a)$$

Example

Find the **average value** of the function $f(x)=x^2+3$ on the interval $[2,5]$

INTEGRATION

1.3

The Fundamental Theorem of Calculus

In this section, we will learn about:

The Fundamental Theorem of Calculus and its significance.

THE MEAN VALUE THEOREM FOR INTEGRALS

Theorem 1.3: The Mean Value Theorem for Integrals

If $f(x)$ is continuous over an interval $[a, b]$, then there is at least one point $c \in [a, b]$ such that

$$f(c) = \frac{1}{b-a}\int_a^b f(x)dx. \tag{1.15}$$

This formula can also be stated as

$$\int_a^b f(x)dx = f(c)(b-a).$$

FUNDAMENTAL THEOREM OF CALCULUS PART 1:
Integrals and Antiderivatives

Theorem 1.4: Fundamental Theorem of Calculus, Part 1

If $f(x)$ is continuous over an interval $[a, b]$, and the function $F(x)$ is defined by

$$F(x) = \int_a^x f(t)dt, \tag{1.16}$$

then $F'(x) = f(x)$ over $[a, b]$.

Generalization

$$\frac{d}{dx}\int_{v(x)}^{u(x)} f(t)dt = u'(x)f(u(x)) - v'(x)f(v(x))$$

Example

If

$$g(x) = \int_1^{x^2} (t^2 - t)dt \quad ,1 \le x$$

Find g’ (x)

$$g(x) = \int_1^{x^2} (t^2 - t)dt$$

$$g'(x) = u'(x) f(u(x)) = 2x(x^4 - x^2), u(x) = x^2$$

FUNDAMENTAL THEOREM OF CALCULUS PART 2:

If f is continuous on $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

where F is any antiderivative of f, that is, a function such that $F' = f$.

INTEGRATION

1.4

Integration Formulas and the Net Change Theorem

In this section, we will learn about:

Integrals with limits that represent

a definite quantity.

NET CHANGE THEOREM

So, we can reformulate as follows.

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

The integral of a rate of change is the net change:

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

NET CHANGE THEOREM

If the mass of a rod measured from the left end to a point x is $m(x)$, then the linear density is $\rho(x) = m'(x)$.

– So,
$$\int_a^b r(x)\,dx = m(b) - m(a)$$
is the mass of the segment of the rod that lies between $x = a$ and $x = b$.

NET CHANGE THEOREM

If the rate of growth of a population is dn/dt, then

$$\int_{t_1}^{t_2} \frac{dn}{dt}\,dt = n(t_2) - n(t_1)$$

is the net change in population during the time period from t_1 to t_2.

- The population increases when births happen and decreases when deaths occur.
- The net change takes into account both births and deaths.

NET CHANGE THEOREM

If an object moves along a straight line with position function $s(t)$, then its velocity is $v(t) = s'(t)$.

– So, $\int_{t_1}^{t_2} v(t)\,dt = s(t_2) - s(t_1)$

is the net change of position, or displacement, of the particle during the time period from t_1 to t_2.

NET CHANGE THEOREM

If we want to calculate the distance the object travels during that time interval, we have to consider the intervals when:

– $v(t) \geq 0$ (the particle moves to the right)

– $v(t) \leq 0$ (the particle moves to the left)

NET CHANGE THEOREM

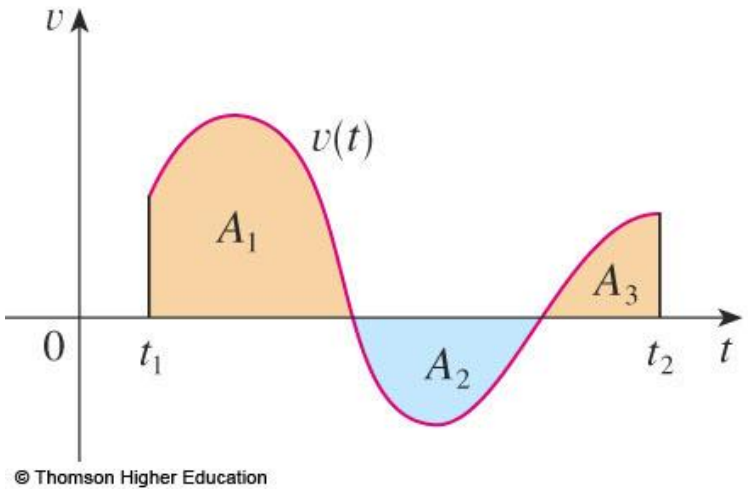
In both cases, the distance is computed by integrating $|v(t)|$, the speed.

Therefore,

$$\int_{t_1}^{t_2} |v(t)|\,dt = \text{total distance traveled}$$

NET CHANGE THEOREM

The figure shows how both displacement and distance traveled can be interpreted in terms of areas under a velocity curve.



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$$\text{displacement} = \int_{t_1}^{t_2} v(t)\,dt = A_1 - A_2 + A_3$$

$$\text{distance} = \int_{t_1}^{t_2} |v(t)|\,dt = A_1 + A_2 + A_3$$

NET CHANGE THEOREM

The acceleration of the object is
$a(t) = v'(t)$.

– So, $\int_{t_1}^{t_2} a(t)\,dt = v(t_2) - v(t_1)$

is the change in velocity from time t_1 to time t_2.

NET CHANGE THEOREM

Example

A particle moves along a line so that its velocity at time t is:

$$v(t) = t^2 - t - 6 \text{ (in meters per second)}$$

a) Find the displacement of the particle during the time period $1 \leq t \leq 4$.

b) Find the distance traveled during this time period.

NET CHANGE THEOREM

Example a

By Equation 2, the displacement is:

$$s(4) - s(1) = \int_1^4 v(t)\,dt = \int_1^4 (t^2 - t - 6)\,dt$$

$$= \left[\frac{t^3}{3} - \frac{t^2}{2} - 6t\right]_1^4 = -\frac{9}{2}$$

– This means that the particle moved 4.5 m toward the left.

NET CHANGE THEOREM

Example b

Note that

$$v(t) = t^2 - t - 6 = (t - 3)(t + 2)$$

– Thus,

$v(t) \leq 0$ on the interval $[1, 3]$ and $v(t) \geq 0$ on $[3, 4]$

NET CHANGE THEOREM

Example b

So, from Equation 3, the distance traveled is:

$$\int_1^4 |v(t)|\,dt = \int_1^3 [-v(t)]\,dt + \int_3^4 v(t)\,dt$$

$$= \int_1^3 (-t^2 + t + 6)\,dt + \int_3^4 (t^2 - t - 6)\,dt$$

$$= \left[-\frac{t^3}{3} + \frac{t^2}{2} + 6t \right]_1^3 + \left[\frac{t^3}{3} - \frac{t^2}{2} - 6t \right]_3^4$$

$$= \frac{61}{6} \approx 10.17 \text{ m}$$

Example

1. Suppose that the animal population is increasing at a rate f(t)=3t-1 (t measured in years).

How much does the animals increase between the third and the seven years?

2. Suppose the acceleration function and initial velocity are $a(t)=t+3$ (m/s^2), $v(0)=5$ (m/s).Find the velocity at time t and the distance traveled when $0<=t<=5$

INTEGRATING OF SYMMETRIC FUNCTIONS

Suppose f is continuous on $[-a, a]$.

a. If f is even, $[f(-x) = f(x)]$, then

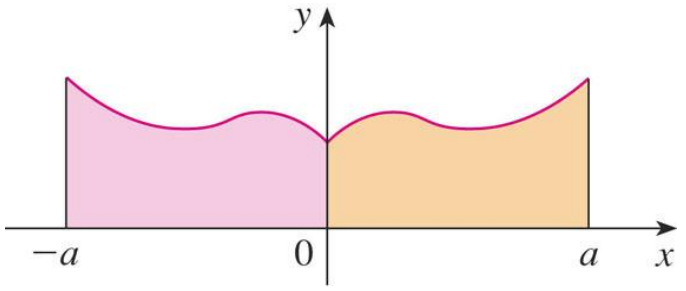
$$\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$$

b. If f is odd, $[f(-x) = -f(x)]$, then

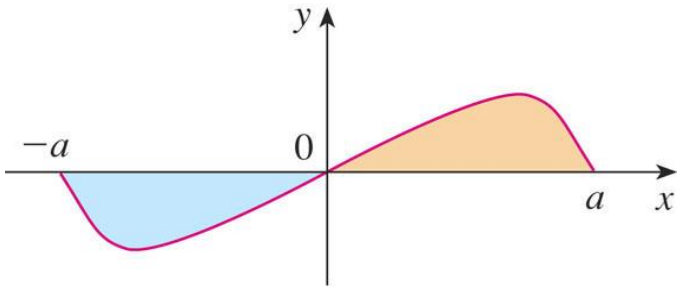
$$\int_{-a}^{a} f(x)\,dx = 0$$

INTEGRATING OF SYMMETRIC FUNCTIONS

This Theorem is illustrated here.



(a) f even, $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$



(b) f odd, $\int_{-a}^{a} f(x)\,dx = 0$

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INTEGRATION

1.5

Substitution

In this section, we will learn:

To substitute a new variable in place of an existing expression in a function, making integration easier.

INDEFINITE INTEGRAL

Due to the relation given by the FTC between antiderivatives and integrals, the notation $\int f(x)\,dx$ is traditionally used for an antiderivative of f and is called an indefinite integral.

Thus, $\int f(x)\,dx = F(x)$ means $F'(x) = f(x)$

INDEFINITE AND DEFINITE INTEGRALS

You should distinguish carefully between definite and indefinite integrals.

– A definite integral $\int_a^b f(x)\,dx$ is a number.

– An indefinite integral $\int f(x)dx$ is a ***function*** (or family of functions).

TABLE OF INDEFINITE INTEGRALS

$$\int cf(x)\,dx = c\int f(x)\,dx$$

$$\int [f(x)+g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

$$\int k\,dx = kx + C$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

$$\int \csc^2 x\,dx = -\cot x + C$$

$$\int \sec x \tan x\,dx = \sec x + C$$

$$\int \csc x \cot x\,dx = -\csc x + C$$

SUBSTITUTION RULE

If $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

INTRODUCTION

Equation 1

However, our antidifferentiation formulas don't tell us how to evaluate integrals such as

$$\int 2x\sqrt{1+x^2}\,dx$$

SUBSTITUTION RULE

Example

Find $\int x^3 \cos(x^4 + 2)\, dx$

- We make the substitution $u = x^4 + 2$.
- This is because its differential is $du = 4x^3\, dx$, which, apart from the constant factor 4, occurs in the integral.

SUBSTITUTION RULE

Example

Thus, using $x^3\,dx = du/4$ and the Substitution Rule, we have:

$$\int x^3 \cos(x^4+2)\,dx = \int \cos u \cdot \tfrac{1}{4}\,du = \tfrac{1}{4}\int \cos u\,du$$

$$= \tfrac{1}{4}\sin u + C$$

$$= \tfrac{1}{4}\sin(x^4+2) + C$$

– Notice that, at the final stage, we had to return to the original variable x.

SUB. RULE FOR DEF. INTEGRALS

Example

Evaluate $\int_1^2 \frac{dx}{(3-5x)^2}$

SUB. RULE FOR DEF. INTEGRALS

Equation 5

If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

Let F be an antiderivative of f.

Then, $F(g(x))$ is an antiderivative of $f(g(x))g'(x)$.

So, $$\int_a^b f(g(x))g'(x)dx = F(g(x))\Big]_a^b = F(g(b)) - F(g(a))$$